

BRAC University (Department of Computer Science and Engineering)
CSE260 Summer 2025

Examination: Quiz 3

Duration: 25 Minutes

Full Marks: 15

Name:

ID:

Section:

Answer the following questions. You **MUST** show your workings/calculations where applicable.

1. Build a calculator that can add or subtract two 4-bit numbers, A and B. The calculator will perform **A-B** if B is **divisible by 4**; otherwise, it will perform **A+B**. Design the circuit using a **4-bit parallel adder-subtractor**. Use external gates if required. [5]

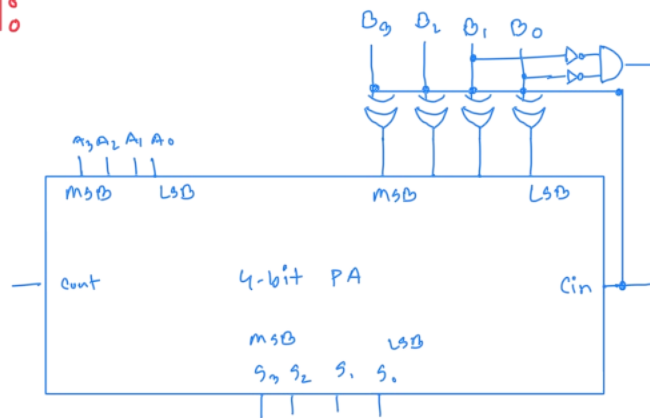
If divisible by 4 $\rightarrow A-B$

else $A+B$

	B_3	B_2	B_1	B_0	$F(\text{div by } 4)$
(0)	0	0	0	0	1
	0	0	0	1	0
	0	0	1	0	0
	0	0	1	1	0
(4)	0	1	0	0	1
	0	1	0	1	0
	0	1	1	0	0
	0	1	1	1	0
(8)	1	0	0	0	1
	1	0	0	1	0
	1	0	1	0	0
	1	0	1	1	0
(12)	1	1	0	0	1
	1	1	0	1	0
	1	1	1	0	0
	1	1	1	1	0

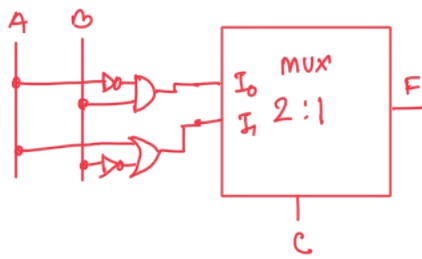
$B_3 B_2$	00	01	11	10
00	1	0	0	0
01	1	0	0	0
11	1	0	0	0
10	1	0	0	0

$\bar{B}_3 \bar{B}_2$



2. Implement the following Boolean function using a single **2:1 MUX**:
 $F(A,B,C) = \sum(1,2,5,7)$. Use external gates if required.

[5]



	I_0	I_1	
$\bar{A}\bar{B}$	0	①	0
$\bar{A}B$	②	3	1
$A\bar{B}$	4	⑤	2
AB	6	⑦	3
$\bar{A}B$	$A + \bar{B}$		
$A\bar{B}$			

3. Implement the following Boolean function using only **2:4 decoder(s)**:

$F(A,B,C) = A \odot B \odot C$. You may use only external **NOR** gates if necessary.

[5]

$$A \odot B = \text{Even } 1$$

Input			Output F	
A	B	C	$A \odot B$	$A \odot B \odot C$
0	0	0	1	0
0	0	1	1	1 $\rightarrow 1$
0	1	0	0	1 $\rightarrow 2$
0	1	1	0	0
1	0	0	0	1 $\rightarrow 4$
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1 $\rightarrow 7$

$$\therefore F = \Sigma(1, 2, 4, 7)$$

